Unified Engineering Workbench

PASS Science Works - R&D Internship 2026-2027 - Technical report draft - 2026-11-01

Candidate: Complete before submission **Implementation:** C++17 local MVP

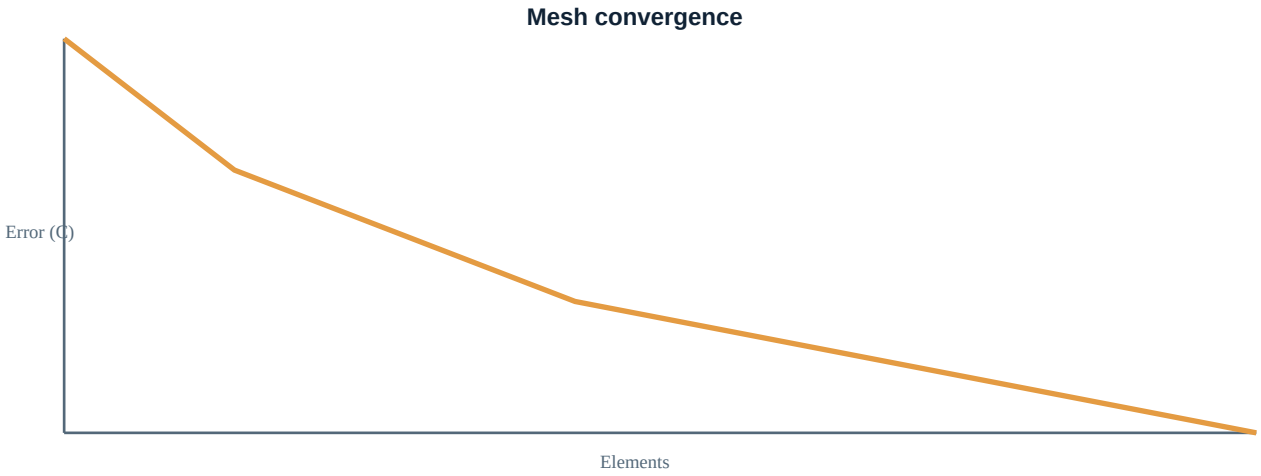
Scope and architecture

This educational MVP combines a C++17 numerical backend and loopback HTTP API with a lightweight browser dashboard. The C++ application implements one-dimensional steady thermal finite-element analysis, a fixed-seed serial/parallel workload, one-neuron backpropagation, and explicitly simulated embedded-device responses. The frontend presents thermal, geometry, performance, hardware, and learning panels; its canvas viewer displays ASCII STL triangles or temperature nodes and supports rotate, pan, zoom, fit, legend, and node inspection.

Thermal method

Assignment defaults: L=1 m, A=1e-4 m^2, k=10 W/(m K), uniform generation q=10000 W/m^3, T(0)=100 C, T(L)=20 C. Each two-node element contributes kA/h [[1,-1],[-1,1]] and consistent load qAh/2 [1,1]. Endpoint temperatures are Dirichlet conditions. Midpoint error compares linear FEM interpolation with the assignment exact quadratic.

Elements	Maximum absolute midpoint error (C)
10	1.25
20	0.3125
40	0.078125
80	0.01953125



Each mesh doubling reduces midpoint interpolation error by four. Nodal values match the exact solution for this constant-property, uniform-source case.

Performance, learning, and limitations

Recorded results are from the local development host and are not portable performance promises.

Serial / parallel benchmark

500 deterministic cases, three solver repeats per case, 20 elements, seed 20261101, and five measurements per configuration. Host: Apple M5, 10 logical CPUs, 16 GiB RAM, Apple Clang 21. Every parallel case was checked against its serial counterpart at 1e-8 tolerance; measured maximum delta was zero.

Workers	Serial median ms	Parallel median ms	Speedup	Efficiency
1	1.30129	1.13933	1.14215	1.14215
2	1.30129	0.937416	1.38817	0.694085
3	1.30129	1.33900	0.971839	0.323946
4	1.30129	0.857750	1.51710	0.379275

Timings are noisy and machine-specific; startup overhead can make a parallel setting slower.

Handwritten backpropagation

For h=tanh(wx+b), yhat=vh+c, and J=0.5(yhat-y)^2, C++ calculates gradients for w, b, v, c and one update at learning rate 0.1. Initial h=0.46211715726, prediction=0.46211715726 and loss=1.182541819. Central finite differences (epsilon 1e-6) gave maximum gradient discrepancy 1.1301892755e-10. The C++ command exports 100 training updates and the loss plot. The candidate must still add the required handwritten scan.

Mandatory delivery risk: no standalone Windows Launch.exe is included or verified on Windows 11. Windows scripts still require CMake/MSVC. Binary STL, quoted CSV parsing, and backend file ingestion remain unsupported. Hardware responses are mocks. Windows verification, saved demo evidence, Git bundle, signed ownership statement, and handwritten appendix remain incomplete. See README.md and docs/STATUS.md. This educational software is not validated engineering design software.